Assignment 2

Name - Neelay Shah
PRN - 123B1B261

Title
Design and Development of a Responsive Personal Portfolio Website Using HTML, Custom CSS, and Bootstrap

Aim
To create a responsive and visually attractive personal portfolio webpage using HTML5, CSS3, and the Bootstrap 5 framework in order to present academic details, work experience, technical skills, and projects effectively.

Objectives
- To organize the structure of the webpage using semantic HTML elements.
- To enhance the appearance of the webpage using custom CSS features such as hover effects, gradient backgrounds, and improved typography.
- To incorporate the Bootstrap 5 framework through a Content Delivery Network (CDN) for faster and easier development without installing files locally.
- To use Bootstrap's 12-column grid system to ensure the layout adjusts smoothly across different devices like mobiles, tablets, and desktops.
- To utilize ready-made Bootstrap components such as navigation bars, cards, and buttons to present information in a clean and structured manner.

Theory

1. HTML5 (HyperText Markup Language)
HTML is the basic language used to build the structure of a webpage. It uses different tags to define elements such as headings, paragraphs, links, and images. In this project, semantic HTML tags are used along with general container elements like and to divide the portfolio into meaningful sections such as Home, About, Experience, Projects, Skills, and Contact.

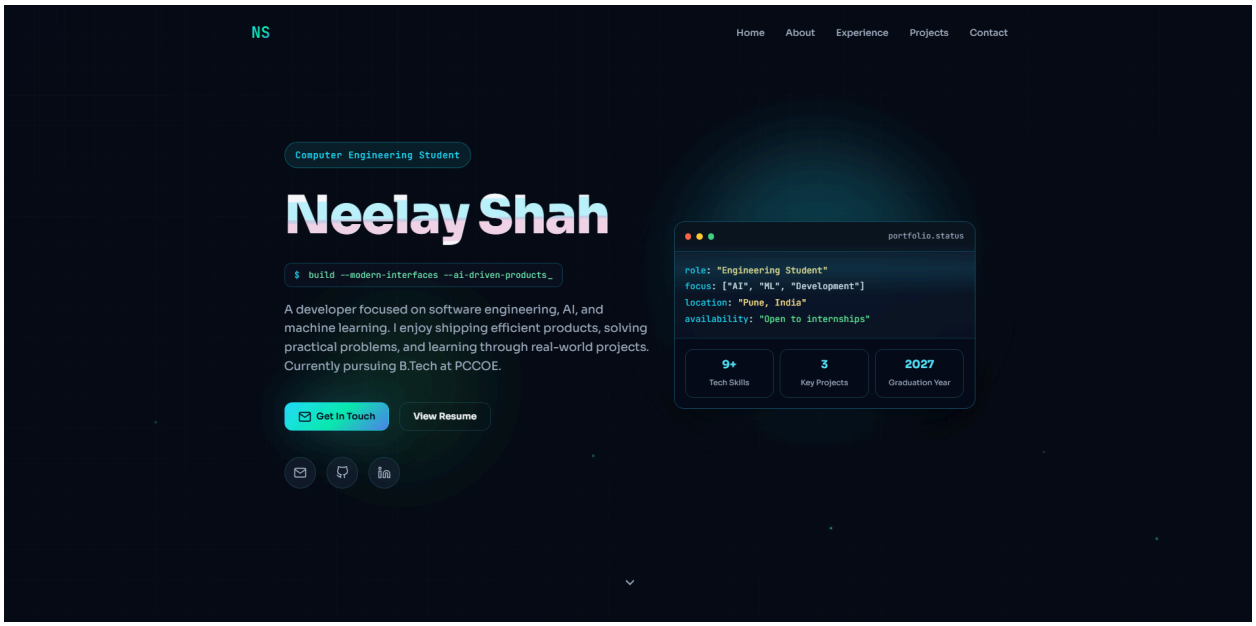
2. CSS3 (Cascading Style Sheets)
CSS is used to control the visual design and layout of the webpage. It manages aspects like colors, fonts, spacing, and overall styling. Although Bootstrap provides built-in styles, custom CSS is applied to give the portfolio a unique look. In this assignment, CSS is added within the <style> tag to implement smooth scrolling, gradient backgrounds for the main section, and interactive effects such as hover animations on project cards.

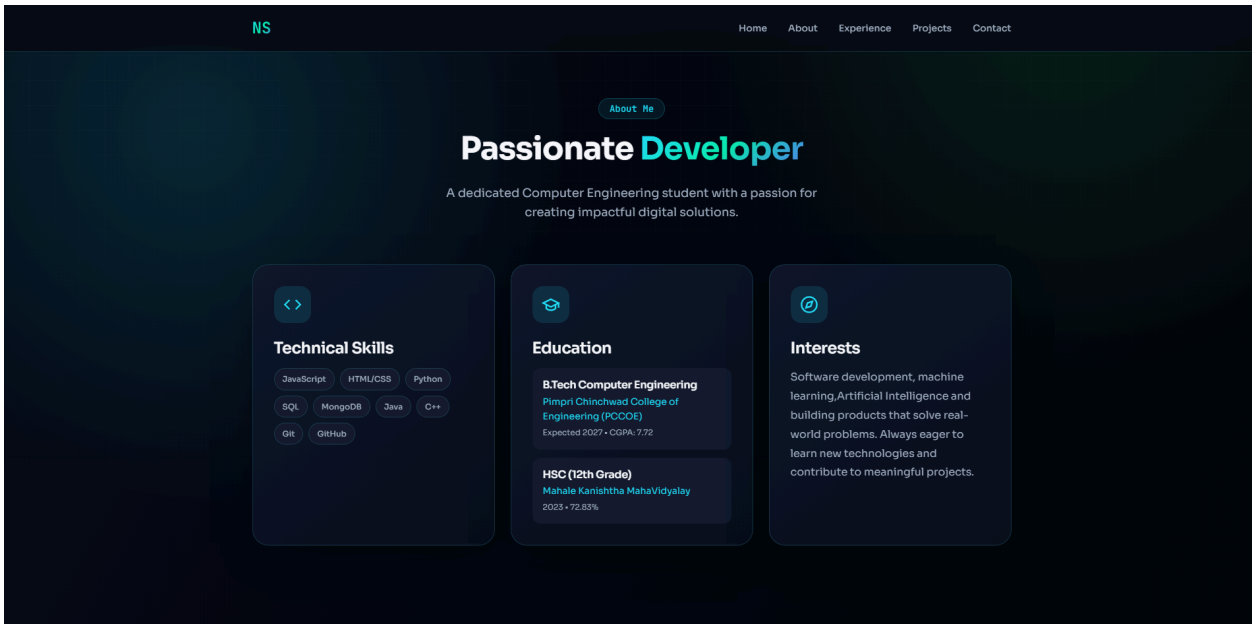
3. Bootstrap 5 Framework and CDN
Bootstrap is a widely used front-end framework that helps in building responsive and mobile-friendly websites efficiently.

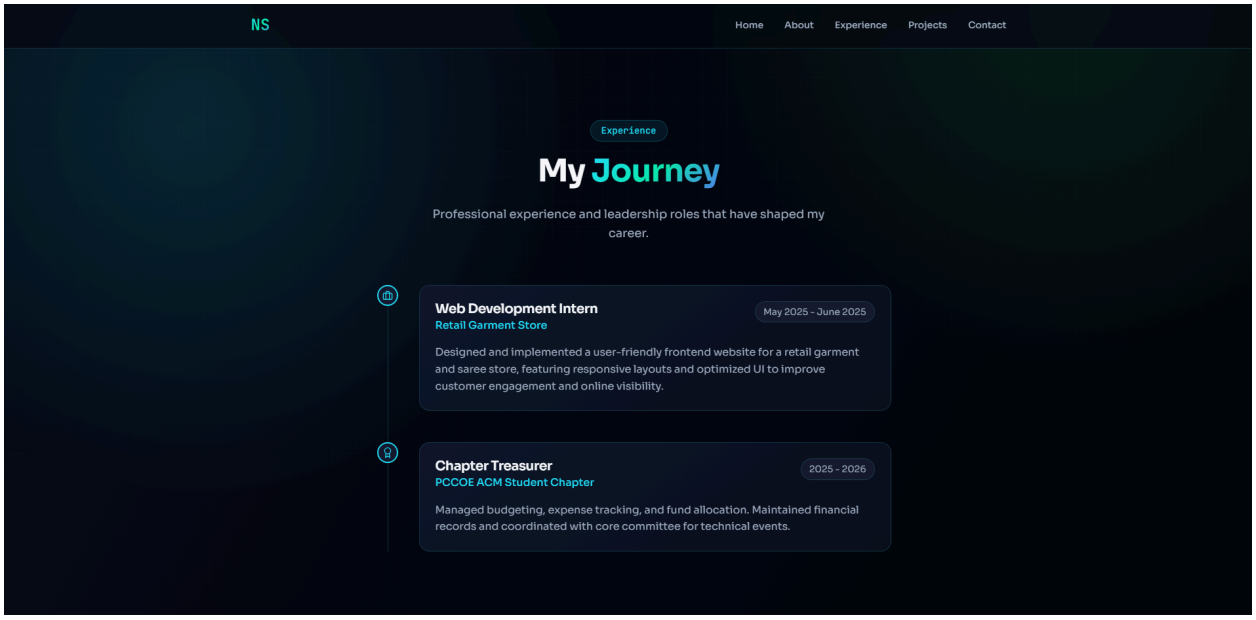
CDN Integration:
Using a CDN allows the webpage to load Bootstrap's CSS and JavaScript files directly from an online server. This removes the need to download files and improves loading speed.
Grid System:
Bootstrap uses a flexible 12-column grid layout based on flexbox. Classes like .col-md-4 and .col-md-6 help in automatically adjusting the layout depending on screen size—for example, stacking elements vertically on smaller screens and aligning them horizontally on larger screens.
Pre-designed Components:
Bootstrap offers ready-to-use components that simplify development. In this assignment, the following are used:

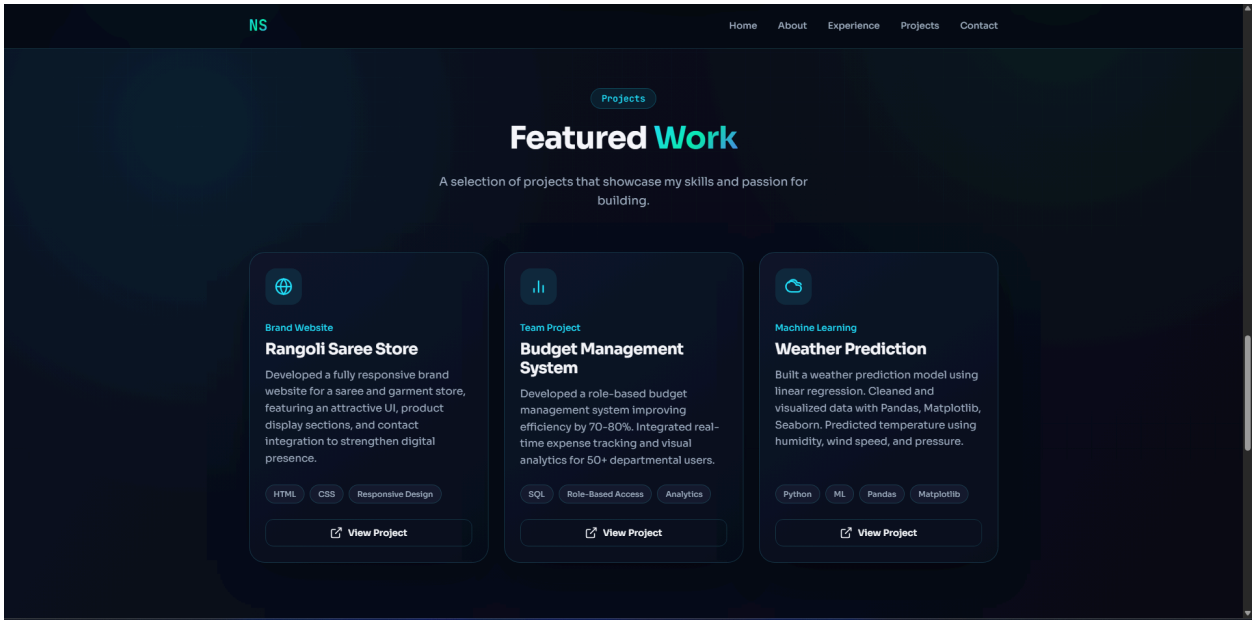
- **Navbar (.navbar):** Creates a responsive header that converts into a collapsible menu on smaller devices.
- **Cards (.card):** Used to display information like projects and experience in a structured box format.
- **Utility Classes:** These are shortcut classes for spacing (like py-5, mb-4), text formatting (text-center, fw-bold), and layout (d-flex), making styling faster without writing extra CSS.

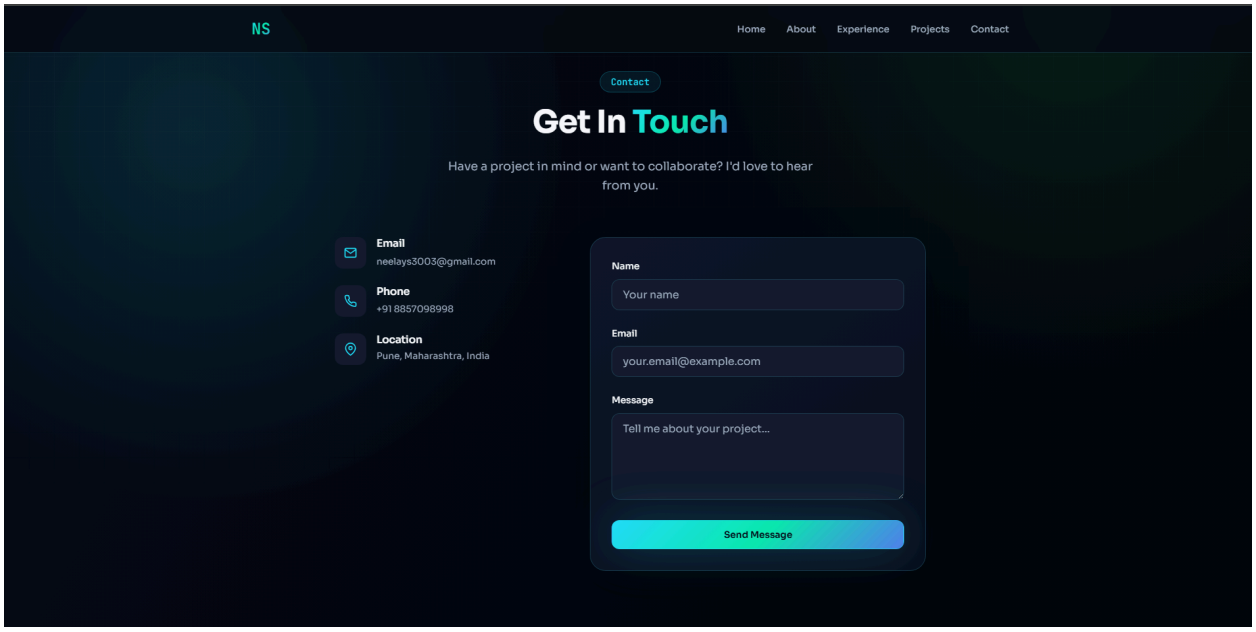
Output -




NS
Home
About
Experience
Projects
Contact
About Me
Passionate Developer
A dedicated Computer Engineering student with a passion for creating impactful digital solutions.
Technical Skills
JavaScript
HTML/CSS
Python
SQL
MongoDB
Java
C++
Git
GitHub
Education
B.Tech Computer Engineering
Pimpri Chinchwad College of Engineering (PCCOE)
Expected 2027 • CGPA: 7.72
HSC (12th Grade)
Mahale Kanishtha MahaVidyalay
2023 • 72.83%
Interests
Software development, machine learning,Artificial Intelligence and building products that solve real-world problems. Always eager to learn new technologies and contribute to meaningful projects.


NS
Home
About
Experience
Projects
Contact
Experience
My Journey
Professional experience and leadership roles that have shaped my career.
Web Development Intern
Retail Garment Store
May 2025 - June 2025
Designed and implemented a user-friendly frontend website for a retail garment and saree store, featuring responsive layouts and optimized UI to improve customer engagement and online visibility.
Chapter Treasurer
PCCOE ACM Student Chapter
2025 - 2026
Managed budgeting, expense tracking, and fund allocation. Maintained financial records and coordinated with core committee for technical events.





Conclusion:

The development of this personal portfolio successfully demonstrates the integration of HTML5, CSS3, and Bootstrap 5 to create a professional and responsive web presence. By leveraging the Bootstrap grid system and utility classes, the application achieves a seamless user experience across various device resolutions, from mobile to desktop. This assignment effectively showcases a diverse technical background—including GenAI deployment, full-stack MERN development, and MLOps pipelines—while maintaining a clean, modern aesthetic through custom styling. Ultimately, the project serves as a functional tool for professional networking, bridging the gap between complex engineering projects and accessible web presentation.